EPSR-test-ex20 nwts: neutron weights of the three EPSR-test samples

Program(s): nwts | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `nwts`, the keyword-driven program that turns the isotopic composition of a sample into the neutron weights of every atom-type pair which `epsr` needs to compare its simulated partial structure factors with a measured differential cross section [1,2]. This case recomputes the three `.wts` files that the EPSR-test refinement actually uses, so it checks the standard single-sample path end to end.

Code exercised

Fortran (original, fortran_src/):

- `nwts.f`: `nwts` (component and isotope loop, exchange averaging, writing `<run>tot.wts`)
- `process_symbols.f`: `check_symbol` (`symbol_setup.eq.'nwts'` branch: `fnameato` sets components, labels and symbols from the `.ato` file); `setup_symbols_generic.f`: `setup_symbols_nwts`
- `read_and_save_files.f`: `read_input_file`, `read_ato` (via `fnameato`), `save_files_others` (rewrites the `.NWTS.dat` file)
- `ato_manipulation_rev4.f`: `scatt_len` (table `sears91_gudrun.inc`); `ato_read_and_save.f`: `minitc`, `moloutc`

Rust (port, rust/epsr-core/src/):

- `tools/nwts.rs`: `run`, `check_symbol_extra`, `scatt_len`, `read_isotopes`; `tools/sears91.rs` (scattering-length table)
- `symbols.rs`: `read_input_file`, `save_inp_file`; `symtab_tools.rs`: keyword table `NWTS`
- `ato.rs`: `read_ato`, `minitc`, `moloutc`

Physical system

The EPSR-test mixture of 200 deuterated dimethyl sulphoxide and 400 water molecules (3200 atoms, 300 K, cubic box 33.12 Å) with six atom types: S, C, O and methyl H of DMSO, O1 and H1 of water. The three samples are d6-DMSO in D2O (H and H1 = 2H), d6-DMSO in H2O (H1 natural) and d6-DMSO in the null-scattering H2O/D2O mixture (H1 = 0.64 natural H + 0.36 2H), the isotope-substitution set of Soper and Luzar [3] measured on ISIS SANDALS. `nwts` looks up the bound coherent scattering length of each isotope [4], averages the abundances, replaces the lengths of exchangeable H1 by the mean over all exchangeable hydrogens, and writes for each pair the intra- and intermolecular products of concentrations and scattering lengths in the Faber-Ziman convention [5] (`output 1`: per atom, `normtot 0`: not normalised).

Inputs

`200DMSO400Water.ato` and `DMSODWater{D,H,Null}.NWTS.dat` copied unchanged from EPSR-test (`nsamples 1`, `iexchange 1` for H1 only); `commands.txt` runs `nwts` once per file.

How the accuracy was verified

`commands.txt` (3 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

The stored reference holds the three files the Fortran writes with new content, `DMSODWater{D,H,Null}.NWTStot.wts` (the rewritten `.NWTS.dat` and `.ato` files come out unchanged). These Fortran weights are themselves byte-identical to the `.NWTStot.wts` files shipped with EPSR-test (checked with `cmp`), so the port reproduces the weights used by the EPSR-test refinement, e.g. 0.0811 for S-S and 0.4417 for C-C of the null sample.

References

[1] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[2] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[4] V. F. Sears, "Neutron scattering lengths and cross sections", Neutron News 3(3), 26-37 (1992).
[5] T. E. Faber and J. M. Ziman, "A theory of the electrical properties of liquid metals III. The resistivity of binary alloys", Phil. Mag. 11, 153-173 (1965).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.